Keerthana Kompella

Curriculum Vitae

Seattle, WA — keerthanakompella5@gmail.com — LinkedIn — GitHub

Education History

BS in Computer Science, Minor in Linguistics, Minor in Mathematics Sept 2023 – June 2027
University of Washington, Seattle, WA
Relevant Coursework: Data Structures, Software Design, Machine Learning, Data Management, Human Computer Interaction, Linear Algebra, Intro to AI, Natural Language Processing, Computer Ethics

Publications

Kompella, K., Chen, A. T. (2026). Documentation as a Coordination Artifact in Undergraduate Research Labs: Reflections from Boundary Object Theory. *To appear in Proceedings of the Association for Information Science and Technology (ASIST 2026).*

Barrett, T., Biira, B., Jawara, J., Shaw, A., Dong, Z., Okolo, C. T., Olojo, S., **Kompella, K.**, Saho, K., Zhang, A. X., Chen, A. T., Battle, L. (2026). Moral Missions: Sustaining Responsible Practice by Surfacing Practitioner Moral Decision-Making Procedures. *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society (AIES 2026)*, 9(1), 349–362. https://doi.org/10.1609/aies.v9i1.43145

Under Review:

Kompella, K., Najjar, N. (2026). A Cognitive Load Theory Perspective on AI-Assisted Design Recipe Environments. *Submitted to the ACM Student Research Competition, SIGCSE Technical Symposium 2027 (SIGCSE TS 2027).* Under review.

Presentations

Kompella, K., Chen, A. T. (Upcoming - 2026, November). Documentation as a Coordination Artifact in Undergraduate Research Labs: Reflections from Boundary Object Theory. Poster accepted for presentation at the Annual Meeting of the Association for Information Science and Technology (ASIST 2026), Thailand.

Kompella, K., Najjar, N. (2026, July). A Cognitive Load Theory Perspective on AI-Assisted Design Recipe Environments. Poster presented at the UNCC Summer Research Symposium, Charlotte, NC.

Kompella, K., Saho, K., Barrett, T., Biira, B., Jawara, J., Shaw, A., Battle, L., & Zhang, A. X. (2026, June). Moral Missions: Prototyping and Designing to Support Moral Decision-Making Procedures of Responsible Data Scientists. Poster presented at the UW Paul G. Allen School Undergraduate and Masters Research Symposium, Seattle, WA.

Kompella, K., Mohamed, S., Trinh, T., Barrett, T., & Battle, L. (2025, June). African Data Ethics: Centering African Values in Artificial Intelligence Resource Page. Poster presented at the UW Paul G. Allen School Undergraduate and Masters Research Symposium, Seattle, WA.

Research Experience

Research Intern May 2026 – Present
Smart & Secure Future Computing REU – UNC Charlotte, Charlotte, NC

- Investigated how Generative AI, integrated with the Design Recipe framework, impacts novice programmers' cognitive load during problem decomposition
- Engineered adaptive scaffolding mechanisms for an AI tool in development using Cognitive Load Theory principles to minimize extraneous load and support iterative problem-solving
- Helped design a HCI lab study utilizing think-aloud protocols and semi-structured interviews to evaluate real-time feedback timing and interactive interface behaviors across students using a new AI education tool
- Synthesized literature across CS Education, learning sciences, and human-AI interaction to ground theoretical considerations, experimental design, and system development

Undergraduate Researcher Jan 2026 – Jun 2026
University of Washington – Moral Missions Project, Seattle, WA

- Conducted qualitative HCI research examining how data science practitioners operationalize personal moral values and ethics into their workflows
- Performed thematic analysis on 41 design goals from 15 interviews with data science practitioners, translating qualitative data into actionable insights for responsible AI tool design

- Rapidly prototyped and facilitated critique sessions for 30+ interactive prototypes targeting practitioner design goals selected and featured in the preprint's appendix

Project Manager and Researcher Apr 2025 – Present
University of Washington – Svoboda Diaries Project, Seattle, WA

- Co-authored research applying Boundary Object Theory to analyze documentation as a coordination artifact in collaborative academic environments
- Served as technical owner and project manager for a distributed digital archival platform, implementing features, managing a web development team, and planning and maintaining long-term system evolution
- Authored and edited documentation, presentation, and data models designed to bridge expertise gaps and lower the cognitive translation burden for future undergraduate maintainers

Lead Developer Dec 2024 – Jun 2025
University of Washington – African Data Ethics Web Tool, Seattle, WA

- Led end-to-end development of a research-driven software tool focused on data ethics, applying modular architecture and user-centered design principles
- Translated abstract responsible AI and ethics research requirements into concrete software features through iterative prototyping and cross-functional testing
- Balanced usability, performance, and maintainability while working with interdisciplinary teams, directly contributing to the African Data Ethics poster research

Employment History

Web Developer Sep 2025 – June 2026
University of Washington – Jackson School of International Studies, Seattle, WA

- Maintained and extended a production web system, collaborating with cross-functional stakeholders to translate requirements into accessible, user-friendly technical solutions
- Debugged and refactored legacy PHP and JavaScript components, reducing system fragility and improving reliability across devices

Full Stack Developer Intern Jun 2025 – Sep 2025
Prime Orbit, Bellevue, WA

- Designed backend services and data pipelines using Java and PostgreSQL, participating in the full software development lifecycle including agile design discussions and code reviews
- Built and maintained data pipelines with PostgreSQL and JDBC, handling validation, concurrency, and failure scenarios

Technology Intern Jun 2024 – Sep 2024
Cascade Counsel, Seattle, WA

- Developed and automated legal technology products and built a functional MVP from an initial business concept to support client demonstrations and workflow efficiency
- Implemented backend automation improvements that reduced manual data entry and improved system efficiency

Teaching Experience

Teaching Assistant, CSE 480 (Computer Ethics) Sep 2026 – Present
University of Washington – Paul G. Allen School of Computer Science & Engineering, Seattle, WA

Teaching Volunteer, 2nd Grade Class Jan 2025 – Mar 2025
Olympic Hills Elementary School, Seattle, WA

Teaching Assistant, AMC 10/12 Camp Jun 2024 – Sep 2024
Art of Problem Solving, Bellevue, WA

Research & Technical Skills

Research Methods: HCI/CSCW Frameworks, Thematic Analysis, Rapid Prototyping, Think-Aloud Protocols, User Testing Scripting, Affinity Diagramming, Literature Reviews, Qualitative Data Analysis
Programming: Java, Python, JavaScript, SQL, C/C++, HTML, CSS
Systems & Tools: React, Node.js, WordPress, Firestore, AWS, Azure, GCP, Git, Git Bash, VS Code, Docker
Program Management: Requirements definition, roadmap planning, milestone tracking, technical documentation, cross-functional collaboration